

LEXIN CHEN

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Computational Chemist | Molecular Modeling & ML for Drug Discovery

Professional Summary

Computational chemist with a Ph.D. in Chemistry and industry experience developing molecular-modeling, conformational-sampling and machine-learning workflows for small molecules and beyond-Rule-of-Five modalities. Experienced in translating structural and ADME data into interpretable compound-design and prioritization strategies.

Skills

Molecular Modeling	Conformational sampling, molecular docking, structure- and ligand-based design, virtual screening, macrocycles and bRo5 modeling, RDKit, Schrödinger Suite, Glide, Maestro, xTB
Cheminformatics/ML	QSAR/QSPR, ADME/property prediction, molecular representations, clustering, scikit-learn, PyTorch, scaffold-aware validation
Programming	Python, Bash, Linux, Git, Docker, AWS, HPC, workflow automation, CI/CD

Experience

Revolution Medicines

Redwood City, CA (Remote)

Scientist I, Computational Chemistry

Dec 2025 - Jul 2026

- Developed and benchmarked a scalable macrocycle conformational-analysis workflow, achieving **<1 Å RMSD against reference crystal structures**.
- Evaluated machine-learning and physics-based sampling approaches, improving conformational accuracy while delivering **3× faster sampling** than baseline methods.
- Analyzed internal Caco-2 permeability data and developed interpretable physicochemical filters and scaffold-aware machine-learning models to enrich for high-permeability compounds and support prioritization.
- Integrated RDKit, Schrödinger, xTB and Python into reproducible molecular-property and conformational-analysis workflows.

Schrödinger

New York, NY

Software Developer Intern

May 2024 - Aug 2024

- Developed and validated algorithms for **Glide**, improving prediction reliability by **15%** and contributing production-quality code to a commercial structure-based drug-design platform.
- Designed docking and molecular-preparation benchmarks across chemically diverse ligands, identifying preprocessing and representation issues affecting model performance.
- Collaborated with scientists, software engineers, and product stakeholders to translate user requirements into tested platform improvements.

University of Florida

Gainesville, FL

Graduate Research Assistant

Oct 2022 - Aug 2025

- Developed **DELIGHT**, an end-to-end ML and cheminformatics framework for learning molecular representations from highly imbalanced DNA-encoded-library screening data, improving recall by **10%**.
- Developed **MDANCE**, enabling billion-scale molecular similarity analysis with a **25× speedup** for dataset curation and ML benchmarking on HPC systems.
- Co-developed **BitBirch**, a scalable clustering algorithm for navigating billion-compound chemical spaces and supporting hit-identification analyses.

Education

Ph.D. in Chemistry

Gainesville, FL

University of Florida

Aug 2025

B.S. in Biochemistry

Stony Brook, NY

Stony Brook University

May 2021

Publication Summary

15 peer-reviewed publications, including 7 first-author papers, in journals such as *J. Chem. Theory Comput.* and *J. Chem. Inf. Model.*, focused on molecular machine learning and scalable cheminformatics.

Selected Honors

Lindau Nobel Young Scientist • Pfizer Emergent Leader • CCG Excellence Award • Penny J. Gilmer Grant